Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Applied AI engineer with **7+ years in real estate and housing SaaS** - the exact domain HomeSmart operates in. At Entrata (100K+ MAU multifamily platform), built Python data pipelines, AI-enabled workflows, and REST API integrations end to end, working directly with internal teams and external stakeholders to take solutions from prototype to production. Now at Intuit, **owning AI solution delivery**: identifying business problems with customers, choosing the right approach (AI vs. deterministic logic), building evaluation harnesses, and monitoring production quality. Built **Leasel** - an 8-agent LangGraph system for housing lease compliance with RAG, document understanding (OCR), human-in-the-loop escalation, and LLM-as-judge evaluation (93.6% precision). Proficient with **LangGraph, LangChain, Python, SQL, APIs, AWS, Docker**, and AI development tools including Claude Code. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python / SQL / REST APIs | LangGraph / LangChain / Orchestration | AI Agents & RAG Pipelines | Document Understanding & OCR | Real Estate & Housing SaaS Domain | Prototype to Production Ownership | LLM Evaluation & Production Quality | Human-in-the-Loop Workflows | AWS (Lambda, S3, RDS, SQS) | Docker / Terraform / GitHub Actions CI/CD | Security & Sensitive Data Handling (PHI/HIPAA) | Data Pipelines & Large-Scale Data Systems | Claude Code / AI Development Tools | Cross-functional Communication

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer - AI Solutions - Virginia, USA

- **Serve as technical point of contact for AI tool vendors and business stakeholders:** scope integration requirements, evaluate third-party APIs, align on data contracts and implementation approach, and drive each initiative from technical discovery through go-live.
- Partner with **TPMs and product managers** to translate ambiguous business requirements into concrete technical solutions; own the delivery timeline and keep stakeholders aligned on tradeoffs and progress.
- Build and deploy **production-ready AI services** in Python/FastAPI on AWS: LLM integrations, retrieval pipelines, structured-output APIs, and guardrails for sensitive financial data handling.
- Build **evaluation harnesses and regression tests** (LLM-as-judge + golden datasets) to validate AI outputs; set expectations with business stakeholders on AI accuracy and edge cases before rollout.

Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer - 100K+ monthly active users

- **Served as primary technical contact for external vendor integrations**: worked directly with HUD and Tax Credit federal data source teams to scope data requirements, debug API and data issues, coordinate go-live timelines, and resolve production incidents end to end.
- Collaborated with **product managers and TPMs** across 120+ Agile sprints to translate client-reported issues into scoped technical solutions; led triage calls and wrote root cause summaries for non-technical stakeholders.
- Delivered **large-scale data pipelines** processing 1.6M+ records with validation and rollback checks; built internal debugging tools that reduced customer issue triage time by 40%.
- Owned **PHI/regulated document workflows**: AWS S3 migration with Libsodium encryption, HIPAA-compliant access controls, and sensitive data handling; optimized 200+ SQL queries (30% gain).
- Built Python/Bash workflow automation eliminating 80% of manual operational work; Docker microservices on AWS with Terraform (50% better scalability); reviewed 300+ pull requests.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer - Pune, India

- Built housing product modules (Python, PostgreSQL) for affordable housing and rental management; delivered large-scale data import and reconciliation workflows (1.6M records).

PROJECTS

Leasel - Multi-Agent AI System for Housing Lease Compliance

Python · LangGraph · FastAPI · RAG · OCR · Human-in-the-Loop

Independently scoped and shipped an AI application for **housing lease compliance** - identifying the business problem, choosing the architecture, and taking it from prototype to a tested production system. 8-agent LangGraph orchestration: triage, document understanding (OCR ingestion), hybrid RAG retrieval, rule evaluation, confidence scoring, and **human review escalation** for low-confidence decisions. LLM-as-judge evaluation harness: 93.6% precision and 87.9% F1 across 1,240 test cases. Security controls route eligibility decisions away from the LLM entirely.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe - Open-Source LLM Evaluation Toolkit

Python · FastAPI · 5 LLM Providers · REST APIs · CI/CD

15-endpoint FastAPI service integrating five AI provider APIs (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama). Evaluates LLM and RAG outputs on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance using LLM-as-judge. CI regression tests with golden datasets. Demonstrates practical AI **evaluation and quality monitoring** in a production-style setup.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017